Cosmic Sort 2026 - Participant Rulebook

Organised by APC Club X NjacK

1. About the event

Cosmic Sort 2026 is a four-day machine learning hackathon on real data from the Sloan Digital Sky Survey (SDSS). Your task is to build a model that works out what an object in the sky is, using only how bright it looks through different colour filters. No astronomy background is needed.

Track	Task	Who
Main track: Star, Galaxy or Quasar?	Classify each object in a table as `STAR`, `GALAXY` or `QSO` (quasar)	Every team. This track decides the overall winners
Bonus track: Galaxy Shapes	Classify galaxy images into 10 shape classes	Optional. Ranked separately

Starter code and a short guide will be published on our GitHub repository before the event.

2. Eligibility and teams

- The event is open to all students. Teams have 1 to 4 members, and each student may join only one team.
- Every member needs their own Kaggle account (free at kaggle.com). Each person may use only one account.
- Team changes are allowed until Day 1, 18:00 IST.

3. Timeline (all times IST)

Date	What happens
Day 0 (29 Oct), 17:00	Kickoff; data released and leaderboards open at 18:00
Day 1, 18:00	Last moment to change teams
Day 4 (1 Nov), 18:00	**Submissions close.** Final submissions are locked

Date	What happens
Day 4, 23:59	Deadline to upload your code on the submission form
Day 6 (3 Nov)	Results announced

4. Data

Main track. A table of real SDSS objects. `train.csv` contains about 73,000 objects with their class, and `test.csv` contains about 31,000 objects to classify. Each object has brightness measurements (magnitudes) in five colour filters (u, g, r, i, z), their uncertainties, a size measurement and its position relative to the Milky Way.

Bonus track. About 14,000 labelled galaxy images and 3,500 test images (128 × 128 pixels). The ten shape classes are: disturbed, merging, round smooth, in-between round smooth, cigar-shaped smooth, barred spiral, unbarred tight spiral, unbarred loose spiral, edge-on without bulge, and edge-on with bulge.

5. Submission

- Submit a CSV file on the track's Kaggle competition page, with one row for every test item and a probability for every class. For example, the main-track header is `object_id,STAR,GALAXY,QSO`, and a row might be `100166,0.97,0.01,0.02`.
- Each team may submit up to **5 times per day** on each track, and must **select one final submission** per track on Kaggle before submissions close.

6. Scoring

Submissions are scored with **log loss**, where lower is better. For every test item, the score looks at the probability you gave to the correct answer: 0.99 costs 0.01, 0.50 costs 0.69, and 0.01 costs 4.61. The penalties are averaged within each class first, so every class counts equally. Being confidently wrong is expensive, so give honest probabilities.

While the event runs, the leaderboard shows scores on **30%** of the test data. The **final ranking** uses the remaining **70%**, which stays hidden until the end. If two teams have the same score, the team that submitted first ranks higher.

7. Rules

1. Make predictions using only the competition data. Do not look up test objects or images in SDSS, Galaxy10 or any other public source, and do not label test items by hand.
2. The `redshift` column appears only in the training data. You may use it to explore the data, but not as an input when predicting the test set.
3. Do not share code or predictions with other teams.
4. Open-source libraries, pretrained models and AI assistants (such as ChatGPT, Claude or Copilot) are allowed.
5. By Day 4, 23:59 IST, every team must upload the code (a notebook) that produced its final submission. A submission without code does not count.
6. Breaking rules 1, 3 or 5, or using more than one Kaggle account per person, leads to disqualification. The organisers' decision is final.

8. Code check

Before results are announced, the organisers re-run the code of the top 10 teams on each track, using only the competition data. Your code must reproduce your final score, allowing for small random differences, and run in under 3 hours on a free Kaggle notebook. If the code cannot reproduce the score, the team is disqualified and the next team moves up.

9. Contact

Avni - 8527587887

Prabhudutta - 7978995595